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Multiclass First-in-First-Out Cell Transmission Model (MF-CTM): A Traffic Flow Model to Estimate Dynamic Road Capacity due to Connected Autonomous Vehicles Operating in Mixed Traffic

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Abstract

Connected Autonomous Vehicle (CAV) is a new technology that can operate without human intervention and able to communicate with other CAVs and connected infrastructure. Due to its ability, CAV is expected to create significant change in traffic management and creates benefits e.g., increasing road traffic performance, road safety improvement. Before CAV has a significant share on the road, there will be a transition period where CAV co-exists with normal vehicle (NV). The purpose of this study is therefore to develop a tool, based on cell transmission model (CTM), to assess the impact of CAV in mixed traffic. We name the model as multiclass first-in-first-out cell transmission model (MF-CTM). The model has three distinct features, namely: i) First-in-First-Out (FIFO), ii) Dynamic Maximum Flow Rate, and iii) Dynamic Maximum Cell Occupancy. In this study, we present the mathematical formulas, algorithm, and numerical illustration of the MF-CTM for a one-lane single road link. By simulation using MF-CTM, we demonstrate that the model can: i) represent decreasing queue length and dissipation time during congestion as CAV share increases, ii) maintain the order of vehicle groups (traffic cohort) during congested condition, iii) represent fluctuation in link outflow rates due to fluctuating CAV shares in the traffic. For further studies, the formulations and algorithm of MF-CTM can be extended into networks and include more elements e.g., signalised intersections, bus lanes, etc. In the future, we expect that the MF-CTM can be a useful tool to assess the effectiveness of traffic management measures involving CAVs.

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1. Introduction

Connected Autonomous Vehicle (CAV) is a new technology in road transport which is expected to bring significant change to traffic operation and management.

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With its ability to operate without human intervention and communicate with other vehicles as well as infrastructure, CAV will be able to travel closer to each other and thus is predicted to improve traffic flow performance and road safety (Ahmed et al., 2022). With the focus on traffic flow, CAV can realise its full benefit to traffic performance improvement when it has dominant share on the road (Mena-Oreja et al., 2019). Before that, there will be a transition period where CAVs co-exist and operate in mixed traffic with normal vehicles (NV). The question is what the impact of CAVs would be to mixed traffic during such transition period. Past studies have been investigating such impact mostly by using microsimulation approach due to its ability to model the traffic in detail (Raju & Farah, 2021) as opposed to macrosimulation. However, microsimulation approach requires a significant amount of computational effort and data at a detailed level. Such requirement may hinder an attempt to model CAV impact on a large-scale network (Levin & Boyles, 2016). Therefore, there is a need to develop an alternative model which represent vehicles as groups (instead of individual vehicle), sufficiently model the reality, and able to represent CAV impact to traffic propagation and queuing.

The aim of this study is to answer the above need by developing a model for CAVs in mixed traffic by using cell transmission model (CTM). The model is called Multiclass FIFO Cell Transmission Model (MF-CTM). The MF-CTM has two contributions to offer. Firstly, this model incorporates first-in-first-out (FIFO) framework by Bar-Gera and Carey (2022) on how vehicles within the CTM proceed from one cell to the next. Vehicle propagation in CTM will depend on whether they follow FIFO or not, which will affect the CAV proportion in each cell. FIFO feature incorporated in the model will make vehicle movement more realistic and CAV proportion in the model more accurate compared to non-FIFO conformant models. A CTM model without FIFO has a high risk of unintended overtaking which might not be possible. Secondly, by using the changing CAV proportion in each cell, the MF-CTM dynamically adjusts the maximum flow rate and maximum cell occupancy in each timestep. Due to CAVs ability to communicate and travel closer to each other, in MF-CTM the cell with higher CAV proportion will have higher value of maximum flow rate and cell occupancy. This is due to CAVs impact in lowering average vehicle reaction time and reducing jam spacing within the cell. As a start, we develop MF-CTM for a one lane-single straight link. By using MF-CTM, we then demonstrated i) the impact of CAV share to queue length and queue dissipation time during traffic congestion, ii) FIFO impact on representing CAV proportion in each cell, iii) impact of CAV share fluctuation to link outflow. This study then explains the mathematical formulations, algorithms and numerical simulations of the MF-CTM. Relevant nomenclatures and notations are summarized below.

Nomenclature and Notations

CAV	: connected autonomous vehicle
NV	: normal vehicle
$CACC$	: cooperative adaptive cruise control
ACC	: adaptive cruise control
HD	: human driving
i	: cell number, for $i = 1$ to I (number of the last cell)
t	: timestep number, for $t = 1$ to T (number of the last timesteps)
v_f	: free flow speed
m	: vehicle type, with $m \in \{1 = CAV, 2 = NV\}$
τ	: traffic cohort order number in a cell
τ^*	: the last traffic cohort order number in a cell
x_i^t	: number of vehicles or cell occupancy in cell i at timestep t
$x_i^{\tau,t}$	: number of vehicles contained in a traffic cohort with order number τ in cell i at timestep t
$x_{a,i}^t$	: number of vehicles in cell i at timestep t that has been adjusted due to the presence of CAV
$x_{i,m}^{\tau,t}$	: number of vehicles in traffic cohort with order number τ in cell i at timestep t for vehicle type m
lp	: the order number (τ) of the last traffic cohort selected to proceed to the next cell at the next timestep
$x_i^{lp1,t}$	: part of traffic cohort $x_i^{lp,t}$ that will proceed to next cell at the next timestep
$x_i^{lp2,t}$	: part of traffic cohort $x_i^{lp,t}$ that will stay at the current cell at the next timestep
u_i^t	: vehicle flows into cell i at timestep t

Q_i^t	: maximum flow rate to cell i at timestep t per timestep
N_i^t	: maximum cell occupancy of cell i at timestep t
w_i^t	: wave speed in cell i at timestep t
$p_i^{\tau,t}$	: the proportion of CAV of cohort τ in cell i at timestep t

2. Literature Review

CTM was firstly developed by Daganzo (1994) which provides formulation for traffic propagation in a single link. In the following year, Daganzo (1995) extends the CTM further by developing formulations for merging and diverging. Since then, CTM has been developed for different context (Adacher & Tiriolo, 2018). The first CTM for CAV in mixed traffic context was developed by Levin & Boyles (2016a) called multiclass CTM. In this CTM extension, traffic in a cell is divided into CAV and NV, and the maximum flow rate to the next cell is adjusted, one of which, based on the proportion of CAV in the current cell. More recently, Qin & Wang (2019) also develop a CTM for CAV called mixed CTM. In this CTM, the traffic is not differentiated into CAV and NV. Instead, the overall CAV proportion in the traffic is used to uniformly adjust the maximum flow rate and maximum cell occupancy for all cells. Although the mixed CTM has simple formulations and assumptions, it has been used by later CTM studies to model CAV in mixed traffic for different context, i.e., link with moving bottleneck (Wu et al., 2022), isolated for-legged junction (Yao et al., 2022), hysteresis in cell fundamental diagram (Jin et al., 2022). Interesting to note that from a CTM formulation for a single link, the formulations become a foundation to model a more complex context e.g., merging/diverging, intersections, etc. Hence, it is important to develop a single link CTM formulations, which in our case is for CAV context, that sufficiently represent the reality. Referring to the two formulations for a single link (Levin & Boyles, 2016a; Qin & Wang, 2019), there are two aspects that can be improved. Firstly, these past studies have not considered first-in-first-out (FIFO), which risk unintentional/unrealistic overtaking and undermine the accuracy and results of the model (Bar-Gera & Carey, 2022). Whether a model consider FIFO or not will also affect the CAV proportions within each cell and having CAV proportions based on a more realistic vehicle movement i.e., FIFO is better than no organising principles. Secondly, each cell capacity (maximum flow rate) and space (maximum cell occupancy) will differ from timestep to timestep based on CAV proportions. The higher the CAV proportion, the higher the maximum flow rate and cell occupancy with an upper bound as CAVs require less space and can travel closer together. Thus, incorporating dynamic maximum flow rate and cell occupancy features in a CTM is important to represent changing road capacity and available space due to changing CAV proportion. However, such aspects have not/or only partially incorporated in the past CTM models for CAV. In a mixed traffic represented by CTM, CAVs may impact other aspects as well such as lane changing or merging traffic (Carey et al., 2015), interaction with traffic signal and bus lane (Shirke et al., 2019), or queue discharge process (Srivastava et al., 2015). However, this study focuses on the most basic features for a single link, namely FIFO, dynamic maximum flow rate and cell occupancy before extending to incorporating complex features further.

3. Methodology

The Multiclass FIFO Cell Transmission Model (MF-CTM) is an extension of the basic CTM (Daganzo, 1994) that incorporates CTM features in Bar-Gera & Carey (2022); Levin & Boyles (2016); Qin & Wang (2019). In a CTM model, the road link is discretised into cells (1 to i) and simulation time into timesteps (0 to t). The illustration of link discretisation into cells in a CTM is shown in Figure 1(a). Vehicle inflows then enter the link (through cell 1) and propagates from one cell to the next (from timestep to timestep) until it exits the link as vehicle outflows. The flows of vehicles (u_i^t) among cells is based on two CTM governing formulas, namely the flow conservation and flow rate formula, consisting of cell parameters and variables namely: free flow speed (v_f), maximum flow rate (Q_i^t), maximum cell occupancy (N_i^t), backward wave speed (w_{i+1}^t), and number of vehicles within the cell (x_i^t). These parameters and variables are derived from traffic triangular fundamental diagram which exists within each cell as shown in Figure 1(b). Traffic performance in a CTM is then measured by number of vehicles within cell. When there is a disruption in the link represented by e.g. significant drop of Q_i^t or N_i^t in one of the cells, queue will form upstream of the disrupted cell(s) represented by increasing number of vehicles in those upstream cell(s). When the disruption is removed, queue

then dissipate and, after some time, the traffic will return to free flow speed condition. The CTMs' ability to represent queue formation and dissipation while modelling vehicles as groups is the reason we select the CTM as the approach for our study. By MF-CTM, we investigate the impact of CAV to mixed traffic by looking at the queue formation and dissipation as well as the parameters and variables surrounding it.

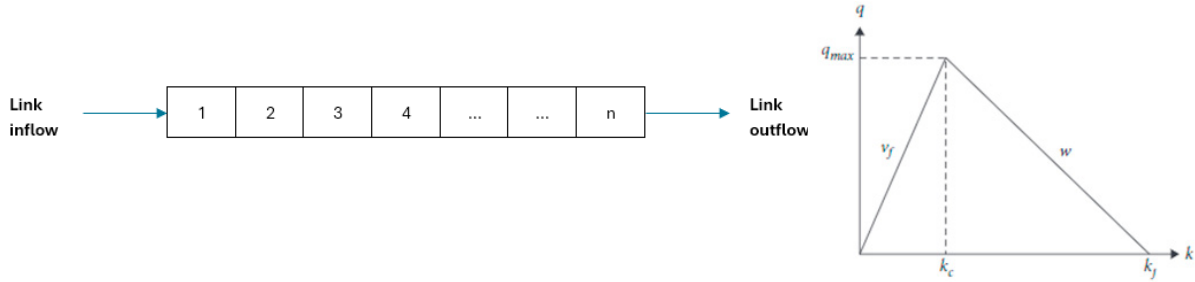


Figure 1. (a) discretisation in a CTM and (b) triangular fundamental diagram which exists within each cell

The MF-CTM is further described as follows. In a basic CTM the number of vehicles in cell i is represented by x_i^t . In MF-CTM, the x_i^t is differentiated further based on its order entering the cell (τ) which we call traffic cohorts ($x_i^{\tau,t}$). The purpose of using traffic cohorts ($x_i^{\tau,t}$) is to ensure that the cohort with higher τ will not overtake the one with lower τ , thus FIFO is adhered, and no unintended or unrealistic overtaking is taking place. Within each cohort, there are CAVs and NVs, represented by $x_{i,m}^{\tau,t}$ as in (1). The use of traffic cohort in MF-CTM then modifies the flow conservation (2) and flow rate formula (3). In brief, the formula (2) describes the number of vehicles in cell i at timestep $t + 1$ (x_i^{t+1}), which is the number of vehicles in cell i at timestep t ($\sum_{\tau=1}^{\tau^*} x_i^{\tau,t}$) added by inflows to cell i and subtracted by outflows to cell $i + 1$. The formula (3) represents the vehicle flows going to cell $i + 1$ (u_{i+1}^t), which is the minimum value of three variables, namely the number of vehicles in cell i , maximum flow rate to cell $i + 1$, and the remaining vehicle spaces in cell $i + 1$ at timestep t .

$$x_i^{\tau,t} = \sum_{m=1}^m x_{i,m}^{\tau,t} \text{ for } m \in \{1 = \text{CAV}, 2 = \text{NV}\} \quad (1)$$

$$x_i^{t+1} = \sum_{\tau=1}^{\tau^*} x_i^{\tau,t} + u_i^t - u_{i+1}^t \quad (2)$$

$$u_{i+1}^t = \min \{ \sum_{\tau=1}^{\tau^*} x_i^{\tau,t}, Q_{i+1}^t, \frac{-w_{i+1}^t}{v_f} (N_{i+1}^t - \sum_{\tau=1}^{\tau^*} x_{a,i+1}^{\tau,t}) \} \quad (3)$$

After we find the amount of flows to cell $i + 1$ at timestep t (u_{i+1}^t) by using equation (3), we select the traffic cohorts in cell i that will be included in u_{i+1}^t based on FIFO. The selection starts from the cohort with the lowest order ($\tau = 1$) until the front part of the last selected cohort ($\tau = lp1$), where $\sum_{\tau=1}^{lp1} x_i^{\tau,t+1} = u_{i+1}^t$. Based on the selected cohorts ($\tau = 1$ to $lp1$), we can then determine the CAV proportion within u_{i+1}^t ($p_{f,i+1}^t$) by equation (4a). Independent of the cohort selection process, we can also determine the CAV proportion within $x_{i+1}^{\tau,t}$ (p_{i+1}^t) by equation (4b).

$$p_{f,i+1}^t = \frac{\sum_{\tau=1}^{lp1} (x_i^{\tau,t} p_i^{\tau,t})}{\sum_{\tau=1}^{lp1} x_i^{\tau,t}} \quad (4a)$$

$$p_{i+1}^t = \frac{\sum_{\tau=1}^{\tau^*} (x_{i+1}^{\tau,t} p_{i+1}^{\tau,t})}{\sum_{\tau=1}^{\tau^*} x_{i+1}^{\tau,t}} \quad (4b)$$

By using $p_{f,i+1}^t$ and p_{i+1}^t , we then adjust the value of the 2nd and 3rd variables of equation (3) respectively. Based on the adjusted variables, we then obtain an extra outflow to be added to u_{i+1}^t (if any) and re-select the traffic cohorts to be included within u_{i+1}^t . We then carry out the flow conservation operation as in (2) to update the number of vehicles within each cell at the next timestep. The simulation can then proceed to the next cell within the same timestep and the next timestep. The idea of the aforementioned process is that the higher the CAV proportion in the traffic, the

higher the maximum flow rate and maximum cell occupancy in the model. This idea is also the reason we incorporate dynamic maximum flow rate and maximum cell occupancy feature in the model, so it can model the fluctuating impact of CAVs to the traffic based on its share. To realise this process, there are many more equations involved. However, it is not shown here due to limited spaces. Nevertheless, the algorithm of for this process is shown in Figure 2.

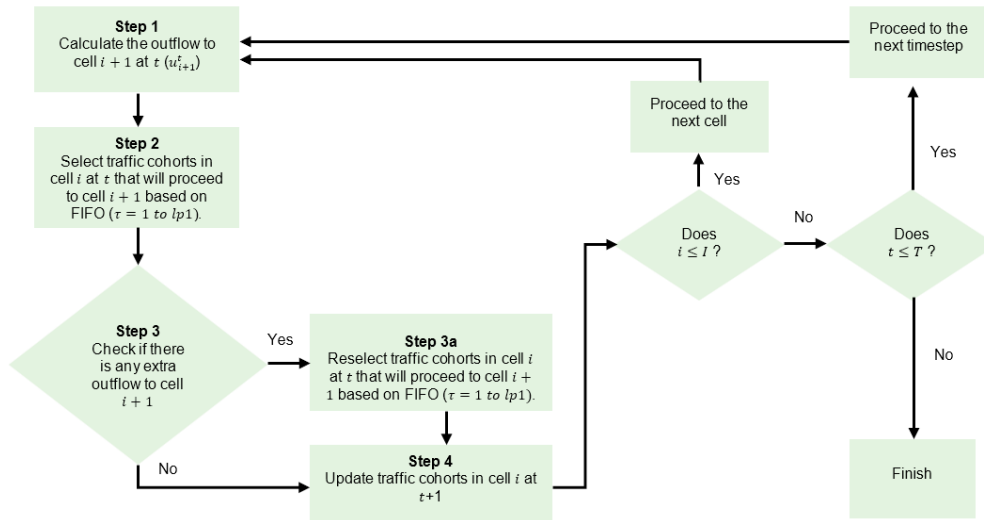


Figure 2. MF-CTM algorithm

4. Numerical Simulation

To illustrate the behaviour of the MF-CTM, this study carries out a numerical simulation for a one-lane single straight link. The aim is two-fold: i) observe the ability of MF-CTM to represent CAVs impact to mixed traffic, particularly on queue formation and dissipation, ii) investigate the significance of incorporating FIFO, dynamic maximum flow rate, and dynamic maximum cell occupancy features into a CTM. For the first aim, we run scenario 1 which set the CAV proportion in the link inflows as constant, and gradually increase the proportion from 0.5, 0.75, and 1. We then visually assess the impact of CAV proportion to the number of congested cells upstream when disruption occurs, and the time it takes to clear the congestion after the disruption is removed. For the second aim, we run scenario 2 which set CAV proportion as random for each timesteps (following uniform distribution) and compare the result with a non-FIFO model namely multiclass CTM by Levin & Boyles (2016). We then observe the CAV proportion in each cell during the simulation and see whether MF-CTM provide a more logical result in terms of FIFO. In addition, we also observe the amount of link outflows and its fluctuation for each timesteps to see whether the dynamic maximum flow rate and cell occupancy makes any significant differences. To accommodate the aforementioned aims and scenarios, this study sets the simulation parameters and variables as follows: the link is discretised into 40 cells and run for 160 timesteps. Starting from timestep 0, inflows amounting at 13.33 veh/timesteps (1600 veh/hr) come into and traverse the link through to the end before exiting the link. At timestep 46, disruption is introduced in the link which makes the maximum flow rate of a cell in the middle of the link (or cell 20) drops to only 25% of its value. The disruption stays for 40 timesteps during the simulation and then removed so the maximum flow rate of the disrupted cell is restored to 100%. The simulation scenarios, parameters and variables can be seen in Table 1 and Table 2.

Table 1. Simulation scenarios

Scenario No	Scenario Name	CAV Proportion in the Link Inflows ($p_{f,1}^t$)	Presence of Disruption	Remaining Maximum Flow Rate of Cell 20 During Disruption
1	Constant CAV proportion	Constant at 0.5, 0.75, 1	Yes	25%
2	Random CAV proportion	Follow uniform distribution		

Table 2. Simulation variables and parameters

Parameter/Variable	Value
Cell length (Δx)	500 m
Timestep length (Δt)	30 s
Number of timesteps	160
Warm up timesteps	0 to 40
Main simulation timesteps	41 to 160
Disruption location	Cell 20 (km 10)
Disruption start timestep	46
Disruption length	1200 sec or 40 timesteps
Free flow speed (v_f)	60 km/h or $16.66 \frac{m}{s}$
Link inflows (u_1^i)	- 1600 veh/hr or 13.33 veh/timesteps for $t = 0$ to 86 - 0 veh/hr or 0 veh/timesteps for $t = 87$ to 160
Reaction time - T_n (s)	0.6 (CACC), 1.1 (ACC), 1.5 (NV)
Jam spacing - d_n (m)	5 (CACC), 6 (ACC), 7 (NV)

This simulation demonstrated that the MF-CTM can:

- Model lower queue length and queue dissipation time resulting from higher CAV proportion in the link inflow. This can be seen in Figure 3 with x-axis: timestep, y-axis: cell (or location in the link), and colour indicator: cell occupancy. In the figure, as the CAV proportion in the link inflow increases from 0.5 to 1, the queue length and dissipation time become lower. Such impact of CAV in mixed traffic to queue length and dissipation time cannot be represented by basic CTM, as assumes the traffic as homogenous and does not outline the formula to achieve dynamic maximum flow rate and maximum cell occupancy. If we use the basic CTM instead to carry out the simulation, then the queue length and dissipation time will be long as it only considers NV.
- Produce a more realistic distribution of CAV proportion among cells, compared to a non-FIFO model. This can be seen in Figure 4 (a) and (b) with x-axis: timestep, y-axis: cell (or location in the link), and colour indicator: CAV proportion in the cell. In the figure, the MF-CTM can maintain the variability of CAV proportion in the cells even after vehicles are mixed together in a queue. This is not the case for the Multiclass (non-FIFO) CTM.
- Achieve fluctuation in the link outflow rate due to the dynamic maximum flow rate and cell occupancy. This is a result of varying CAV proportion as explained by point 2) above. This can be seen in Figure 4 (c) with x-axis: timestep, y-axis: outflow per timestep. We can see that due to varying CAV proportion in the link outflow, the value of the link outflow also varies. Such variation cannot be captured by the Multiclass (non-FIFO) CTM.

5. Conclusion

This study develops a model to assess Connected Autonomous Vehicle (CAV) impact to road traffic flow based on Cell Transmission Model (CTM). The motivation is to develop a model that represents vehicles as groups, so it does not require data at a detailed level, while also able to sufficiently model traffic flow propagation and queuing. The model is called Multiclass FIFO Cell Transmission Model (MF-CTM). MF-CTM advances past CTM models for CAV by i) incorporating first-in-first-out feature to increase the accuracy of the model, ii) incorporating dynamic maximum flow rate and iii) maximum cell occupancy to represent CAV impact (based on CAV fluctuating share or proportion), to the fluctuating of road capacity, available space, and link outflow. Through numerical simulation, we show that the MF-CTM is able to i) model decreasing queue length and dissipation time as CAV share or proportion increases, ii) maintain the order of vehicles (traffic cohorts) with their unique CAV proportion even during congested condition, iii) model fluctuating link outflow due to fluctuating CAV proportion. The development of MF-CTM in this study is still in its early phase, in which in which we only develop a model for a one-lane single straight link.

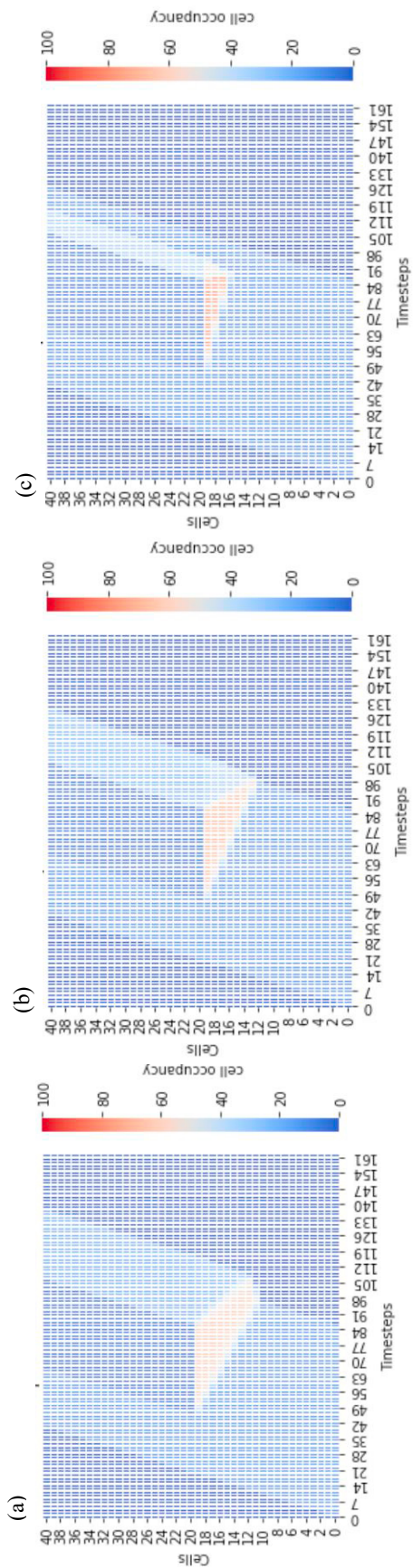


Figure 3. Simulation results of the MF_CTM: Cell occupancy in Scenario 1 with (a) $p_{f,1}^c = 0.5$, (b) $p_{f,1}^c = 0.75$, (c) $p_{f,1}^c = 1$

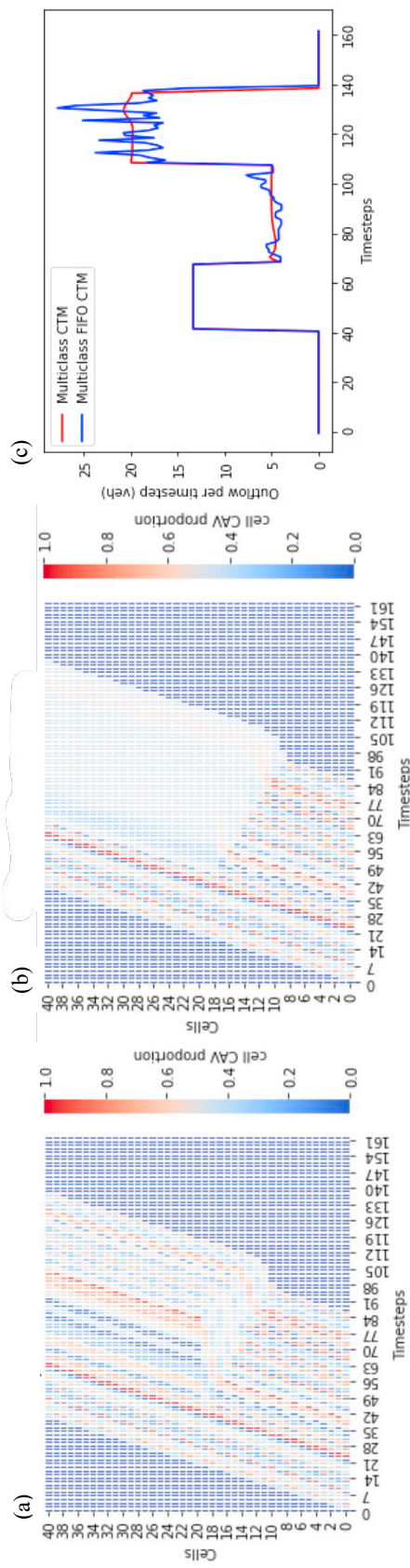


Figure 4. Simulation result - CAV proportion within cells comparison (scenario 2) for (a) MF_CTM, (b) Multiclass CTM and (c) link outflow per timestep

For further research, MF-CTM can be expanded for network traffic by adding formulas for merging and diverging. More elements can also be added into the model e.g., signalized intersection, bus lanes, etc. After those further development, we expect MF-CTM can be used to test policies related to managing mixed traffic with CAVs e.g., traffic signal priority, signal optimization, bus lane with intermittent priority, etc.

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